

HATS-2b

R-2293

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATS-2_210316 · created 2026-08-17T01:04:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459289.8224 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.04
Transit depth (Rp/Rs)^2	1.02 +/- 0.81 [%]
Orbital Inclination (inc)	85.12 +/- 2.01
Airmass coefficient 1 (a1)	1.194 +/- 0.022
Airmass coefficient 2 (a2)	-0.09 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#1 - [464, 93]
Optimal Method	PSF photometry
Transit Duration (day)	0.047 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.101 $\pm$ 0.04 vs 0.1335 (deviation 0.81 σ)
Duration fit vs published	0.047 d vs 0.0871 d (deviation 8.49 σ)
Mid-time O–C	6.8 min
Depth SNR	2.5
Residual scatter	1.82 %

Light curve

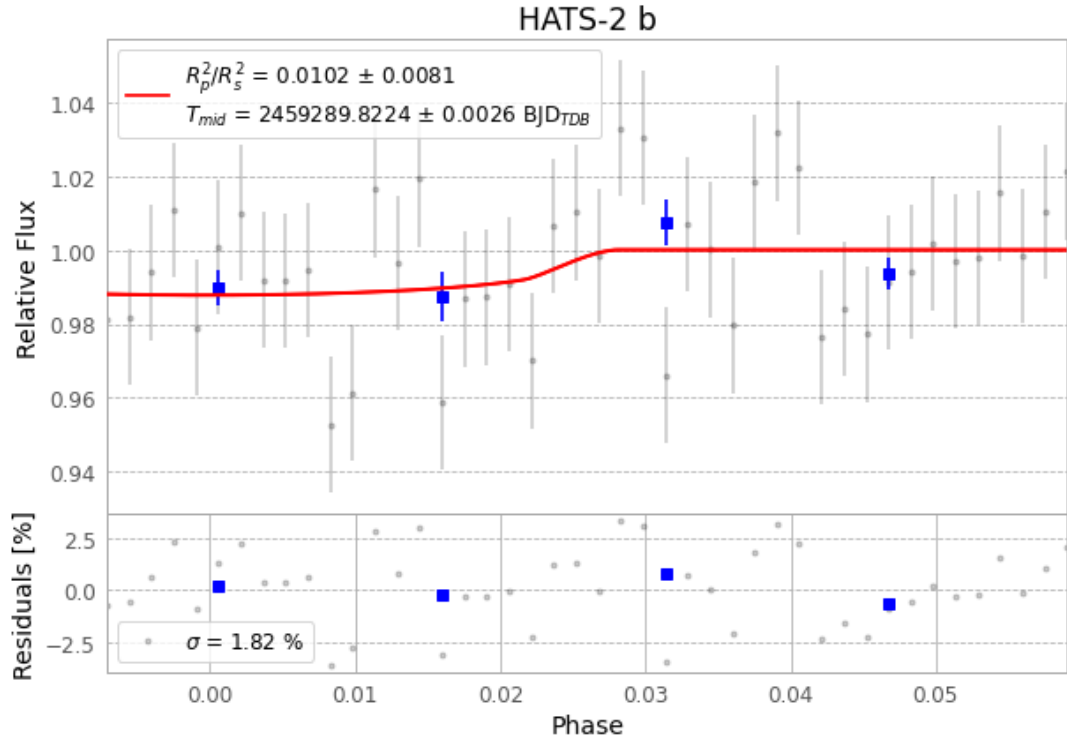


Figure 1 — FinalLightCurve_HATS-2 b_2021-03-16.png (SHA-256 3b71ef7138547903...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 182], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a3bf991cb2693d19...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

44 FITS frames (29 MB), HATS-2210316072112.FITS → HATS-2210316093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATS-2-210316.log (07adc33dac30...), FinalLightCurve_HATS-2 b_2021-03-16.png (3b71ef713854...), FinalLightCurve_HATS-2 b_2021-03-16.csv (5da7801cb260...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 01:04Z.